

Erol OREL, PhD, FRM

Econometrician - Data Scientist - Financial Risk Manager

Data Science

Proficient in data wrangling, feature engineering and NLP to optimize datasets for analysis. Competence in unsupervised machine learning techniques such as dimensionality reduction and clustering to uncover meaningful patterns. Expertise in econometrics and supervised machine learning models with proven ability in training, testing, and prediction scoring methods. Comfortable with concepts such as feature selection, class imbalances, cross-validation and hyperparameter tuning to improve model performance.

Finance

Strong foundation in Arbitrage Pricing Theory and multi-factors modeling and proficiency in risk monitoring and reporting, ensuring comprehensive oversight of investment portfolios. Skilled in stress testing, scenarios, and sensitivity analysis to assess portfolio resilience in diverse market conditions. Ability to strategically choose alternative funds to formulate effective investment strategies. Well-versed in GIPS performance attribution measurement and reporting. Knowledge in legal and compliance matters.

Work Experience

Senior Research Associate - Infectious Diseases and Mathematical Modelling

Institute of Global Health - Campus Biotech - Geneva - 2019 / current

Understanding infectious diseases epidemic by identifying and integrating relevant risk factors using supervised and unsupervised learning algorithms with Python and R. Prediction of individual HIV status using gradient boosted trees in Python. Development of a web application ([LiteRev](#)) for streamlining literature review of scientific papers using NLP, dimension reduction, clustering and nearest neighbors search in Python. [COVID-19 epidemiological dashboard](#) in the Africa region using R Markdown for WHO.

Cofounder / Data Scientist - Asset Management Services

Market Intuitive Investment Managers - Geneva - 2014 / 2018

Arbitrage Pricing Theory and multi-factors modeling using econometric models. Time-varying risk premia and alternative risk premia estimation. Active monitoring and portfolio analysis to assist clients in undertaking suitable performance oversight. Advisory on mutual and alternative investment funds (long-short equity, statistical arbitrage, global macro, CTA) selection. Production of performance and risk reports considering investment objectives, risk levels and returns relative to target returns.

Financial Risk Manager / Senior Quantitative Analyst - Asset Management Services

Steenman Asset Management - Geneva - 2007 / 2014

Asset allocation and investment analysis

Construction of qualitative and quantitative asset allocation strategies for multi-asset class portfolios with a substantial proportion in alternative investments. Development of systematic portfolio management tools and rebalancing methods based on investment objectives and specified risk levels. Analysis, clustering, comparison and selection of mutual funds and alternative investment funds such as long-short equity, statistical arbitrage, global macro, and CTA.

Risk management and performance analysis

Selection and measurement of financial risk metrics such as Value at Risk and expected shortfall for non-gaussian distribution using extreme value theory as a complement to existing models. Time-series modeling using risk premia and alternative risk premia in R. Dynamic tracking of risk exposures, implementation of stress tests, macroeconomic scenarios and sensitivity analysis. Performance attribution analysis and measurement on different scales.

Project Management

Establishment of three alternative investment vehicles (SIF/SICAV) based in Luxembourg. Integration of Webfolio back-office platform for enhanced shadow accounting. Implementation of an automated GIPS Compliance client reporting factsheet in VBA.

Responsible for Research - Cantonal Service of Statistical Research and Information

SCRIS - Lausanne - 2007

Teaching Experience

Lecturer - Statistics / Epidemiology

MAS in Public Health / Faculty of Medicine - University of Geneva - 2024 / current

Research Assistant - Economic Indicators of the Geneva Economy LEA-PICTET-OCSTAT

Laboratory of Applied Economics (LEA) - Geneva - 2005 / 2006

Teaching Assistant - Statistical Methods (Bachelor) / Statistical Modeling (Master of Science)

Faculty of Economics - University of Geneva - 2005 / 2006

Education

PhD in Biomedical Sciences - Global Health Track - Faculty of Medicine

University of Geneva - 2019 / 2023

Dissertation title: « Enhancing HIV Prediction through Machine Learning and Streamlining Research with LiteRev »

I-Teams Medical Innovation Program - Faculty of Medicine

University of Geneva - 2023

Financial Risk Manager - FRM® program

Certified by the Global Association of Risk Professionals (GARP) - 2018

Master of Advanced Studies in Econometrics (D.E.A.) - Faculty of Economics

University of Geneva - 2004 / 2006

Dissertation title: « Additive Models and Local Kernel M-Estimator: Bandwidth Selection and Application to the Implied Volatility Surface »

Master of Science in Quantitative Economics (Licence) - Faculty of Economics

University of Geneva - 2000 / 2004

Most Relevant Publications ([Google Scholar](#))

Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. "An Automated Literature Review Tool (LiteRev) for Streamlining and Accelerating Research Using Natural Language Processing and Machine Learning: Descriptive Performance Evaluation Study." J Med Internet Res (2023); 25: e39736.

Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. "Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa." PLoS ONE (2022); 17(3): e0264429.

Language and Computer

French: Mother tongue - **English:** Fluent - **Turkish:** Father tongue - **German:** Basic - **Spanish:** Basic

Programming Languages and Statistical Software: Python, R, SQL, Git, Bash, Slurm, Matlab, S-Plus, SPSS, Stata, SAS, Eviews

Financial Software and Office: Bloomberg, Datastream, BarraOne, FactSet, Webfolio, Pertrac, VBA, Power BI, Tableau

Interests

Tennis, Chess, Golf

List of Publications

Oral Communications

“Welcome to LiteRev, Streamlining Literature Reviews”. BioTechX Conference Europe. Switzerland, 2023.

“Welcome to LiteRev, Streamlining Literature Reviews”. Colloque de l’ISG. Switzerland, 2023.

“LiteRev: An Automated Literature Review Tool for Streamlining and Accelerating Research”. IWHOD, Poster Session. Greece, 2023.

“Prediction of Individual HIV Status based on Socio-Behavioural Characteristics and Mapping of the Literature”. HUG Colloque HIV. Switzerland, 2022.

“Topic Modelling of Scientific Articles: An Example using CORD-19 Database”. SIMED Lab Meeting. Switzerland, 2022.

“Topic Modelling of Scientific Articles: An Example using CORD-19 Database”. Colloques du G6. Switzerland, 2021.

“Topic Modelling of Scientific Articles: An Example using CORD-19 Database”. UNIGE Data Science Day. Switzerland, 2021.

“Predicting Individual HIV Status using Social and Behavioural Factors in Eastern and Southern Africa”. Microbe Interaction Retreat. Switzerland, 2019.

“Predicting Individual HIV Status using Social and Behavioural Factors in Eastern and Southern Africa”. CODATA/DDI Alliance Dagstuhl Workshop. Germany, 2019.

List of Publications

Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. “Difference in clinical severity between Delta variant and Omicron sub-lineages in a Swiss tertiary center.” CMI communications (2024).

Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. “An Automated Literature Review Tool (LiteRev) for Streamlining and Accelerating Research Using Natural Language Processing and Machine Learning: Descriptive Performance Evaluation Study.” J Med Internet Res (2023); 25: e39736.

Esra R, Carstens J, Le Roux S, Mabuto T, Eisenstein M, Keiser O, Orel E, Merzouki A, De Voux L, Maskew M, Sharpey-Schafer K. “Validation and Improvement of a Machine Learning Model to Predict Interruptions in Antiretroviral Treatment in South Africa.” J Acquir Immune Defic Syndr (2023); 92(1):42-49.

Ng'ambi WF, Merzouki FA, Estill J, Orel E, Chimbandule T, Nyirenda R, Keiser O. “Factors associated with the risk of HIV infection among HIV-exposed infants in Malawi: 2013-2020.” BMJ Paediatr Open (2022); 6(1): e001275.

Botero-Mesa S, Coelho F, Nwosu K, Wicht B, Venkatasubramanian A, Wagner O, Valera C, et al. “Leveraging human resources for outbreak analysis: lessons from an international collaboration to support the sub-Saharan African COVID-19 response.” BMC Public Health (2022); 22(1): 1073.

Ng'ambi W, Estill J, Jahn A, Orel E, Chimbandule T, Nyirenda R, Keiser O. “Factors associated with HIV viral suppression among children and adults receiving antiretroviral therapy in Malawi in 2021: Evidence from the Laboratory Management Information System.” Tropical Medicine & International Health (2022); 27(7): 639-646.

Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. "Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa." PLoS ONE (2022); 17(3): e0264429.

Ng'ambi W, Merzouki A, Estill J, Orel E, Chimbandule T, Nyirenda R, Keiser O. "Factors associated with the risk of HIV infection among HIV-exposed infants in Malawi: 2013–2020." BMJ Paediatrics Open (2022); 6(1).

Nwosu K, Fokam J, Wanda F, Mama L, Orel E, Ray N, Meke J, et al. "SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon." Nature Communications (2021); 12(1): 5851.

Estill J, Islam T, Houben R, Rudman J, Ragonnet R, McBryde E, Trauer J, et al. "Tuberculosis in the Western Pacific Region: Estimating the burden of disease and return on investment 2020–2030 in four countries." The Lancet Regional Health–Western Pacific (2021); 11.

Merzouki A, Estill J, Orel E, Tal K, Keiser O. "Clusters of sub-Saharan African countries based on sociobehavioural characteristics and associated HIV incidence." PeerJ (2021); 9: e10660.

Mboussou F, Impouma B, Farham B, Wolfe C, Williams G, Ngom R, Nzingou M, et al. "Analysing the reported incidence of COVID-19 and factors associated in the World Health Organization African region as of 31 December 2020." Epidemiology & Infection (2021); 149: e256.

Merzouki A, Styles A, Estill J, Orel E, Baranczuk Z, Petrie K, Keiser O. "Identifying groups of people with similar sociobehavioural characteristics in Malawi to inform HIV interventions: a latent class analysis." Journal of the International AIDS Society (2020); 23(9): e25615.

Thiabaud A, Triulzi I, Orel E, Tal K, Keiser O. "Social, behavioral, and cultural factors of HIV in Malawi: semi-automated systematic review." Journal of medical internet research (2020); 22(8): e18747.

Estill J, Venkova-Marchevska P, Roelens M, Orel E, Temerev A, Flahault A, Keiser O. "Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model." F1000Research (2020); 9.